

Answers From Chatgpt

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1 Explanation of the $SU(2)$ Global Anomaly Remark

We explain the logic of the remark step by step.

1.1 $SU(2)$ Gauge Theory in Four Dimensions

The weak interaction is described by an $SU(2)$ gauge theory. Therefore, on our four-dimensional spacetime M there is an $SU(2)$ principal bundle

$$P \rightarrow M.$$

On a local patch we may treat spacetime as $\mathbb{R}^4$, and a gauge transformation is a map

$$g : \mathbb{R}^4 \rightarrow SU(2).$$

1.2 Compactification to S^4 and Large Gauge Transformations

Assume the gauge transformation becomes trivial at spatial infinity:

$$g(x) \rightarrow \mathbf{1} \quad \text{as } |x| \rightarrow \infty.$$

This allows us to compactify $\mathbb{R}^4$ by adding a point at infinity:

$$\mathbb{R}^4 \cup \{\infty\} \cong S^4,$$

with $g(\infty) = \mathbf{1}$. Hence gauge transformations are classified by maps

$$g : S^4 \rightarrow SU(2) \simeq S^3.$$

The homotopy classes of such maps are elements of

$$\pi_4(S^3) \simeq \mathbb{Z}_2 = \{0, 1\}.$$

The nontrivial class (1) corresponds to a “large” gauge transformation.

1.3 SU(2) Doublet Fermions

Fermions charged under the weak force transform in the fundamental (doublet) representation $\mathbf{2}$ of $SU(2)$. A single such fermion field is a section of the associated vector bundle

$$\mathbf{2} \otimes S \longrightarrow M,$$

where S is the spinor bundle. For our purposes, the only important fact is:

A single SU(2) doublet fermion is one section of a vector bundle.

1.4 Witten's SU(2) Global Anomaly

A key result of Witten (1982) states that under a large gauge transformation $g : S^4 \rightarrow S^3$, the partition function of a single $SU(2)$ doublet fermion transforms as

$$Z_{\text{single doublet}} \longrightarrow (-1)^s Z_{\text{single doublet}},$$

where $s = [g] \in \pi_4(S^3) \simeq \mathbb{Z}_2$. Thus

$$s = 0 : Z \text{ invariant}, \quad s = 1 : Z \mapsto -Z.$$

Therefore:

A single SU(2) doublet suffers from a global SU(2) anomaly.

1.5 Multiple Doublets and Anomaly Cancellation

If there are N doublets, the total transformation is

$$Z \longrightarrow (-1)^{Ns} Z.$$

The theory is gauge invariant for all gauge transformations if and only if N is even. In the Standard Model, one generation contains

$$3 \text{ quark doublets} + 1 \text{ lepton doublet} = 4 \text{ doublets}.$$

Hence

$$(-1)^{4s} = +1,$$

and the anomaly cancels.

This explains, for instance, why one cannot simply change $SU(3)$ color to $SU(4)$: it would change the number of weak doublets, potentially making N odd and producing an inconsistency.

1.6 Summary

- Large $SU(2)$ gauge transformations are classified by $\pi_4(S^3) \simeq \mathbb{Z}_2$.
- A single doublet fermion changes sign under the nontrivial transformation.
- A consistent theory must have an *even* number of doublets.
- The Standard Model satisfies this automatically ($N = 4$ per generation).

2 Fiber Choice in the Covering Map $\mathbb{R} \rightarrow S^1$

Consider the covering map

$$p : \mathbb{R} \longrightarrow S^1, \quad p(x) = e^{2\pi i x}.$$

For any point $z = e^{2\pi i \alpha} \in S^1$, the fiber is

$$p^{-1}(z) = \{x \in \mathbb{R} \mid e^{2\pi i x} = z\} = \alpha + \mathbb{Z},$$

a translated copy of $\mathbb{Z}$. Thus every fiber is homeomorphic to the discrete set $\mathbb{Z}$.

In the long exact sequence of a fibration, one chooses a basepoint $b_0 \in S^1$ and defines the fiber as $p^{-1}(b_0)$. Although $b_0 = 1 \in S^1$ (the identity element) is the conventional choice, any b_0 works because S^1 is path-connected and homotopy groups based at different points are naturally isomorphic.

3 Homotopy Groups of the Discrete Space $\mathbb{Z}$

Although $\mathbb{Z}$ is not a manifold, it is a valid topological space and its homotopy groups are well-defined.

3.1 All Continuous Maps $S^n \rightarrow \mathbb{Z}$ Are Constant ($n \geq 1$)

Since $\mathbb{Z}$ is discrete, each singleton $\{k\} \subset \mathbb{Z}$ is open. A continuous map from a connected space S^n to a discrete space must have connected image, hence the image consists of only one point. Thus any map

$$f : S^n \rightarrow \mathbb{Z}, \quad n \geq 1$$

is constant.

3.2 Computation of $\pi_n(\mathbb{Z})$

If we fix the basepoint $k_0 \in \mathbb{Z}$, the only based map $S^n \rightarrow \mathbb{Z}$ is the constant map $f(x) \equiv k_0$. Hence, for $n \geq 1$,

$$\pi_n(\mathbb{Z}, k_0) = 0.$$

For $n = 0$, the path components of $\mathbb{Z}$ are just its points, so

$$\pi_0(\mathbb{Z}) \cong \mathbb{Z}.$$

3.3 Summary

$$\boxed{\pi_0(\mathbb{Z}) = \mathbb{Z}, \quad \pi_{n \geq 1}(\mathbb{Z}) = 0.}$$

3.4 Application: Computing $\pi_n(S^1)$ Using the Covering Space

From the fibration

$$\mathbb{Z} \longrightarrow \mathbb{R} \longrightarrow S^1,$$

the long exact sequence gives

$$\pi_n(S^1) \cong \pi_{n-1}(\mathbb{Z}).$$

Using our calculation of $\pi_{n-1}(\mathbb{Z})$, we obtain the classical result:

$$\pi_n(S^1) = \begin{cases} \mathbb{Z}, & n = 1, \\ 0, & n \neq 1. \end{cases}$$

4 Understanding why $\partial : \mathbb{Z} \rightarrow \mathbb{Z}$ must be multiplication by 2

Consider the fibration

$$SO(2) \longrightarrow SO(3) \longrightarrow S^2.$$

The long exact sequence in homotopy (for $n = 3$ in the book's notation) contains the segment

$$\pi_2(SO(2)) \longrightarrow \pi_2(SO(3)) \longrightarrow \pi_2(S^2) \xrightarrow{\partial} \pi_1(SO(2)) \longrightarrow \pi_1(SO(3)) \longrightarrow \pi_1(S^2).$$

We substitute the known homotopy groups:

$$\pi_2(SO(2)) = 0, \quad \pi_2(S^2) = \mathbb{Z}, \quad \pi_1(SO(2)) = \mathbb{Z}, \quad \pi_1(SO(3)) = \mathbb{Z}_2, \quad \pi_1(S^2) = 0.$$

Thus the exact sequence becomes

$$0 \longrightarrow \pi_2(SO(3)) \longrightarrow \mathbb{Z} \xrightarrow{\partial} \mathbb{Z} \xrightarrow{\text{mod } 2} \mathbb{Z}_2 \longrightarrow 0.$$

4.1 Step 1: Determining the form of ∂

Exactness at $\pi_1(SO(2)) = \mathbb{Z}$ means

$$\text{Im}(\partial) = \ker(\text{mod } 2).$$

Since

$$\ker(\text{mod } 2) = 2\mathbb{Z},$$

we obtain

$$\text{Im}(\partial) = 2\mathbb{Z}.$$

Any group homomorphism $\partial : \mathbb{Z} \rightarrow \mathbb{Z}$ must have the form

$$\partial(n) = kn, \quad k \in \mathbb{Z}.$$

Its image is $k\mathbb{Z}$. To satisfy $\text{Im}(\partial) = 2\mathbb{Z}$, the only possibility is

$$k = \pm 2.$$

Hence

$$\partial(n) = \pm 2n.$$

This is the meaning of the sentence in the book:

“This forces $\partial : \mathbb{Z} \rightarrow \mathbb{Z}$ to be a multiplication by 2.”

4.2 Step 2: Computing $\pi_2(SO(3))$

Exactness at $\pi_2(S^2) = \mathbb{Z}$ implies

$$\text{Im}(\pi_2(SO(3)) \rightarrow \mathbb{Z}) = \ker(\partial).$$

Since $\partial(n) = \pm 2n$, its kernel is

$$\ker(\partial) = \{0\}.$$

Thus

$$\text{Im}(\pi_2(SO(3)) \rightarrow \mathbb{Z}) = 0.$$

But the map $0 \rightarrow \pi_2(SO(3))$ at the beginning of the sequence is injective, so the only way $\pi_2(SO(3))$ can map to 0 inside $\mathbb{Z}$ is if

$$\pi_2(SO(3)) = 0.$$

So the conclusion is:

$$\boxed{\pi_2(SO(3)) = 0.}$$

5 Unitarity in 2-2 scattering

The partial-wave coefficients are defined by projecting the T -matrix onto two-particle states of definite total momentum and total angular momentum,

$$f_J(p^2) \propto \frac{\langle p, J, \vec{m} | \hat{T} | p, J, \vec{m} \rangle}{\langle p, J, \vec{m} | p, J, \vec{m} \rangle}. \quad (1)$$

Because $\hat{T}$ is invariant under translations and under $SO(1, d-1)$, the matrix element in this basis is diagonal in $(J, \vec{m})$, so the proportionality constant is independent of p^2 and of $\vec{m}$. The result therefore depends only on the Lorentz-invariant combination $p^2 = s$.

We start from the angular form of the elastic unitarity equation,

$$2 \Im T(s, t(z)) = \int_{-1}^1 dz' w(z') \int_{-1}^1 dz'' w(z'') \mathcal{P}_d(z, z', z'') T(s, t(z')) T^*(s, t(z'')), \quad (2)$$

where

$$w(z) = (1 - z^2)^{\frac{d-4}{2}}.$$

Next we insert the partial-wave expansion,

$$T(s, t(z)) = \sum_{L=0}^{\infty} n_L^{(d)} f_L(s) P_L^{(d)}(z), \quad (3)$$

and substitute this expression into (2). At this stage we obtain

$$\begin{aligned} 2 \Im T(s, t(z)) &= \sum_{L, L'} n_L^{(d)} n_{L'}^{(d)} f_L(s) f_{L'}^*(s) \int_{-1}^1 dz' w(z') \mathcal{P}_d(z, z', z'') P_L^{(d)}(z') \\ &\quad \times \int_{-1}^1 dz'' w(z'') P_{L'}^{(d)}(z''). \end{aligned} \quad (4)$$

To isolate the coefficient $f_J(s)$ we project both sides of (4) onto the J -th partial wave, multiplying by

$$\frac{\mathcal{N}_d}{2} w(z) P_J^{(d)}(z)$$

and integrating over z . Using the projection formula

$$f_J(s) = \frac{\mathcal{N}_d}{2} \int_{-1}^1 dz w(z) P_J^{(d)}(z) T(s, t(z)), \quad (5)$$

the left-hand side becomes simply

$$\frac{\mathcal{N}_d}{2} \int_{-1}^1 dz w(z) P_J^{(d)}(z) 2 \Im T(s, t(z)) = 2 \Im f_J(s).$$

Now we insert the kernel decomposition (eq. (67))

$$\mathcal{P}_d(z, z', z'') = \frac{1}{2} \mathcal{N}_d^2 w(z') w(z'') \sum_{L=0}^{\infty} n_L^{(d)} P_L^{(d)}(z) P_L^{(d)}(z') P_L^{(d)}(z''), \quad (6)$$

and use the orthogonality relation

$$\frac{1}{2} \int_{-1}^1 dz w(z) P_J^{(d)}(z) P_L^{(d)}(z) = \frac{\delta_{JL}}{\mathcal{N}_d n_J^{(d)}}. \quad (7)$$

Substituting (6) into (4) and performing the two z', z'' integrals, only the term $L = L' = J$ survives due to (7), giving

$$\frac{\mathcal{N}_d}{2} \int dz w(z) P_J^{(d)}(z) (\text{RHS}) = \frac{(s - 4m^2)^{\frac{d-3}{2}}}{\sqrt{s}} |f_J(s)|^2.$$

Equating both sides yields the elastic partial-wave unitarity condition

$$2 \Im f_J(s) = \frac{(s - 4m^2)^{\frac{d-3}{2}}}{\sqrt{s}} |f_J(s)|^2, \tag{8}$$

which is eq. (68) of the main text.

6 Solution Structure of S_i from Given Helicity Data

We are given three helicities

$$h_1, h_2, h_3 \in \frac{1}{2}\mathbb{Z}, \quad h_1 + h_2 + h_3 > 0, \quad (9)$$

three spins

$$S_i \geq 0$$

(half-integers), and integers

$$k_i \geq 0$$

satisfying

$$S_i - k_i = h_i, \quad i = 1, 2, 3. \quad (10)$$

We also impose the constraint

$$k_1 + k_2 + k_3 \leq 2S_{\min}, \quad S_{\min} := \min(S_1, S_2, S_3). \quad (11)$$

Elementary implications

From (10) and $k_i \geq 0$,

$$S_i = h_i + k_i \Rightarrow S_i \geq h_i. \quad (12)$$

Hence every spin satisfies the effective lower bound

$$S_i \geq \max(0, h_i). \quad (13)$$

Let the ordered helicities be

$$h_{(1)} \leq h_{(2)} \leq h_{(3)}. \quad (14)$$

Then

$$S_{\min} = \min_i S_i \geq \min_i \max(0, h_i) = \max(0, h_{(1)}). \quad (15)$$

Rewriting the constraint

Using $k_i = S_i - h_i$ in (11),

$$k_1 + k_2 + k_3 = (S_1 + S_2 + S_3) - (h_1 + h_2 + h_3) \leq 2S_{\min}. \quad (16)$$

From $S_i \geq S_{\min}$, we have

$$k_i = S_i - h_i \geq S_{\min} - h_i \quad (17)$$

and at the same time $k_i \geq 0$, so we obtain

$$k_i \geq \max\{0, S_{\min} - h_i\} \quad (18)$$

Define

$$f(S_{\min}) := \sum_{i=1}^3 \max\{0, S_{\min} - h_i\}. \quad (19)$$

This quantity is the minimal possible value of $k_1 + k_2 + k_3$ at fixed $S_{\min}$.

Thus a necessary and sufficient condition for feasible k_i is

$$f(S_{\min}) \leq 2S_{\min}. \quad (20)$$

With ordered helicities,

$$f(S_{\min}) = \begin{cases} 0, & 0 \leq S_{\min} \leq h_{(1)}, \\ S_{\min} - h_{(1)}, & h_{(1)} \leq S_{\min} \leq h_{(2)}, \\ 2S_{\min} - (h_{(1)} + h_{(2)}), & h_{(2)} \leq S_{\min} \leq h_{(3)}, \\ 3S_{\min} - (h_1 + h_2 + h_3), & S_{\min} \geq h_{(3)}. \end{cases} \quad (21)$$

Imposing (20) yields:

- Region I:

$$0 \leq S_{\min} \leq h_{(1)}$$

Always satisfied.

- Region II:

$$h_{(1)} \leq S_{\min} \leq h_{(2)}$$

$$S_{\min} - h_{(1)} \geq 0, \quad S_{\min} - h_{(2)} \leq 0, \quad S_{\min} - h_{(3)} \leq 0 \quad (22)$$

So we have

$$f(S_{\min}) = S_{\min} - h_{(1)} \leq 2S_{\min} \quad \Rightarrow \quad S_{\min} \geq -h_{(1)}. \quad (23)$$

If $h_{(1)} < 0$, it will give us a non-trivial lower bound.

- Region III:

$$h_{(2)} \leq S_{\min} \leq h_{(3)}$$

$$S_{\min} - h_{(1)} \geq 0, \quad S_{\min} - h_{(2)} \geq 0, \quad S_{\min} - h_{(3)} \leq 0 \quad (24)$$

So we have

$$f(S_{\min}) = 2S_{\min} - (h_{(1)} + h_{(2)}) \leq 2S_{\min} \quad \Rightarrow \quad h_{(1)} + h_{(2)} \geq 0. \quad (25)$$

- Region IV:

$$S_{\min} \geq h_{(3)}$$

$$3S_{\min} - (h_1 + h_2 + h_3) \leq 2S_{\min} \quad \Rightarrow \quad S_{\min} \leq h_1 + h_2 + h_3. \quad (26)$$

Combining (15) with the former constraints, a solution exists iff

$$h_{(1)} + h_{(2)} \geq 0, \quad (27)$$

and the interval

$$S_{\min} \in \left[\max(0, h_{(1)}), h_1 + h_2 + h_3 \right] \quad (28)$$

is nonempty.

Since

$$h_1 + h_2 + h_3 > 0,$$

the upper bound is always positive.

Thus (27) is the essential obstruction that rules out forbidden helicity configurations.

Whenever a valid $S_{\min}$ exists, the spins are reconstructed by

$$S_i = h_i + k_i, \quad k_i \in \mathbb{Z}_{\geq 0}, \quad (29)$$

with

$$k_i \geq \max(0, S_{\min} - h_i), \quad k_1 + k_2 + k_3 \leq 2S_{\min}. \quad (30)$$

Because both

$$S_{\min}$$

and

$$k_1 + k_2 + k_3$$

are bounded, only finitely many triples

$$(S_1, S_2, S_3)$$

are possible for specific helicities.

6.1 A general lower bound valid for all solutions for $h_1 < 0, h_{2,3} \geq 0$

We do not assume which S_i equals $S_{\min}$. We only use the relations

$$S_i = h_i + k_i, \quad k_i \geq 0, \quad (31)$$

and the definition of the minimal spin

$$S_i \geq S_{\min}. \quad (32)$$

From this inequality we immediately obtain

$$S_{\min} - h_i \leq S_i - h_i = k_i, \quad (33)$$

and therefore

$$k_i \geq S_{\min} - h_i. \quad (34)$$

Combining with $k_i \geq 0$ gives the unified inequality

$$k_i \geq \max\{0, S_{\min} - h_i\}, \quad i = 1, 2, 3. \quad (35)$$

Summing these three inequalities yields

$$k_1 + k_2 + k_3 \geq \sum_{i=1}^3 \max\{0, S_{\min} - h_i\}. \quad (36)$$

Now consider the case of exactly one negative helicity and two positive helicities. After reordering,

$$h_1 < 0 < h_2 \leq h_3. \quad (37)$$

Since each $S_i \geq 0$, we necessarily have

$$S_{\min} \geq 0. \quad (38)$$

We determine the signs of the terms in the previous inequality:

- For $i = 1$: since $h_1 < 0$ and $S_{\min} \geq 0$, we have

$$\max(0, S_{\min} - h_1) = S_{\min} - h_1 > 0. \quad (39)$$

- For $i = 2, 3$: since $h_2, h_3 > 0$, we only need the trivial bounds

$$\max(0, S_{\min} - h_2) \geq 0, \quad \max(0, S_{\min} - h_3) \geq 0. \quad (40)$$

Therefore,

$$k_1 + k_2 + k_3 \geq (S_{\min} - h_1) + 0 + 0 = S_{\min} - h_1. \quad (41)$$

Combining this with the fundamental inequality

$$k_1 + k_2 + k_3 \leq 2S_{\min}, \quad (42)$$

we obtain the chain

$$S_{\min} - h_1 \leq k_1 + k_2 + k_3 \leq 2S_{\min} \implies S_{\min} - h_1 \leq 2S_{\min}. \quad (43)$$

Rearranging gives the universal lower bound

$$-h_1 \leq S_{\min} \implies S_{\min} \geq |h_1|. \quad (44)$$

Since $S_1 \geq S_{\min}$ for all solutions, we also have

$$S_1 \geq S_{\min} \geq |h_1|. \quad (45)$$

Thus we have established the following conclusion:

For any admissible solution, we necessarily have $S_1 \geq |h_1|$.

(46)

This bound required no assumption about which spin equals $S_{\min}$; it is a universal constraint following only from the defining inequalities.

Existence of a solution saturating the lower bound

Assume in addition the necessary helicity condition

$$h_1 + h_2 \geq 0 \quad \Longleftrightarrow \quad h_2 \geq |h_1|, \quad (47)$$

which is required for the existence of any solution.

We now construct an explicit solution saturating the inequality $S_1 \geq |h_1|$. Define

$$S_1 = |h_1|, \quad S_2 = h_2, \quad S_3 = h_3. \quad (48)$$

Using $S_i = h_i + k_i$ we obtain the corresponding integers

$$k_1 = S_1 - h_1 = |h_1| - h_1 = 2|h_1|, \quad k_2 = 0, \quad k_3 = 0. \quad (49)$$

Since $h_2 \geq |h_1|$ and $h_3 \geq h_2$, we have

$$S_2 = h_2 \geq |h_1|, \quad S_3 = h_3 \geq h_2 \geq |h_1|, \quad (50)$$

so that in this configuration

$$S_{\min} = S_1 = |h_1|. \quad (51)$$

Finally,

$$k_1 + k_2 + k_3 = 2|h_1| = 2S_{\min}, \quad (52)$$

which saturates the inequality $k_1 + k_2 + k_3 \leq 2S_{\min}$.

Thus the lower bound is sharp: the minimal possible value of S_1 is

$$\boxed{S_1^{\min} = |h_1|.} \quad (53)$$

An explicit solution attaining this minimum is

$$(S_1, S_2, S_3) = (|h_1|, h_2, h_3), \quad (k_1, k_2, k_3) = (2|h_1|, 0, 0). \quad (54)$$

This completes the demonstration that, in the case of one negative helicity and two positive helicities with $h_2 \geq |h_1|$, we not only have the universal bound $S_1 \geq |h_1|$ but also a concrete solution saturating it.